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Leveraging GraphSAGE and large language models with cross-attention for transaction fraud detection: Evidence from credit card and PaySim datasets☆

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ABSTRACT

Online purchasing fraud has grown with e-commerce, causing economic losses and undermining consumer trust. Due to its diversity and dynamism, few studies have examined e-commerce fraud detection. Traditional machine learning algorithms, such as random forest and support vector machines, have achieved poor performance because they do not always capture the rich contextual metadata or the underlying relational structure between transactions. In response to these shortcomings, we introduce a new hybrid fraud detection method that utilizes a lightweight large language model (specifically, DistilBERT) and Graph Neural Networks (GraphSAGE). Using a cross-attention mechanism, our model learns both structural information encoded in the transaction network and semantic information (e.g., transaction type and metadata). By including rich metadata embeddings and transaction graph structures into its architecture, the model learns to effectively detect frauds, even when signals are weak or unclear. With this connection, fraud detection becomes more efficient and resilient. Two benchmark datasets were used for the experiments: the PaySim financial transaction simulator and the European Credit Card Dataset (ECCD). The suggested model obtained 89.1% accuracy, 85.0% F1-score, and 85.5% AUC on ECCD and 90.0% accuracy, 87.0% F1-score, and 85.0% AUC on PaySim using a chronological 70/10/20 train–validation–test split. These findings show that, in comparison to single-modality models, merging structural transaction graphs with semantic metadata greatly enhances fraud detection performance.

1. Introduction

Credit cards (CCs) have changed the face of international trade and consumer banking in this age of rapid technological advancement. Apart from being basic tools for making payments, they also provide ease, the ability to buy things, and freedom from financial dependence (Baisholan et al., 2025). If used responsibly, credit cards can help raise credit ratings and open doors to better lending terms in the future.

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Online transactions are made more convenient by technical breakthroughs, but these same advancements also make sophisticated fraud possible. Online platforms are being used more and more by cybercriminals to carry out bogus transactions that seem very much like real ones. Global losses from credit and debit card theft surpassed $33.8 billion in 2023, and by 2033, they are projected to top $400 billion, according to the Nilson Report (The Nilson Report, 2025). More than $275 million was misappropriated in the US in the year 2024. These numbers highlight how critical it is to have sophisticated fraud detection systems that can consistently spot sophisticated attempts.

Many other ML and DL methods have been suggested for use in this area of fraud detection. Several methods have been used in the past, such as decision trees, logistic regression (LR), Support Vector Machines (SVM), and decision trees (credit, multilingual), among others. Hidden Markov Models (HMM), neural networks (NN) (Baisholan et al., 2025; Mohawesh, Liu et al., 2023), Random Forests (RF), and Bayesian belief networks (BBN) (Baisholan et al., 2025) show enhanced functioning. To tackle data imbalance and high dimensionality, researchers have also investigated One-Class Classification (OCC) approaches based on subspace learning (Zaffar et al., 2023). However, static or rule-based models are not very good at spotting fraud because apparently acceptable actions frequently mask harmful activity.

According to Liu et al. (2022) (Liu & Yang, 2021), Graph Neural Networks (GNNs) like GraphSAGE are very good at simulating the network architecture of financial transactions since they depict interactions as graph structures. To extract contextual information from transaction metadata, transformer-based models from Natural Language Processing (NLP) like BERT and DistilBERT have been successful (Abu-Salih et al., 2025; Zhang et al., 2025). While natural language processing (NLP) models take relational structures into account, graph-based models pay more attention to topological patterns and less to transaction semantics. When fraud signals are vague or disjointed, both systems fail.

Our proposed hybrid fraud detection method combines GraphSAGE and DistilBERT through a cross-attention technique to align graph structures with transaction semantics, thus overcoming this constraint. The model improves its fraud detection capabilities, particularly in the presence of weak or confusing signals, by combining transaction graphs with extensive information embeddings. A more all-encompassing solution for financial fraud detection, this combination of structural and semantic analysis gets above the constraints of current ML-based methods.

The main contributions of this paper are as follows:

- We introduce a hybrid fraud detection framework combining GraphSAGE and DistilBERT through a cross-attention mechanism.
- We integrate transaction graph structures and rich metadata embeddings, improving detection even with weak or confusing signals.
- We evaluate the framework on two real-world financial datasets using metrics such as area under the curve (AUC), F1-score, recall, accuracy, and confusion matrices.

The purpose of this study is to:

- Develop a hybrid fraud detection framework that integrates transformer-based language models with graph neural networks (GNNs).
- Investigate whether fraud detection accuracy can be improved through multimodal fusion of structural transaction graphs and semantic transaction metadata.
- Evaluate the robustness and generalization capability of the proposed model using both large-scale synthetic financial transaction datasets (PaySim) and real-world credit card transaction datasets (ECCD).

This paper is organized as follows for the rest of it. With a focus on ML and graph-based approaches, Section 2 examines relevant literature on fraud detection. Section 3 describes the hybrid framework that has been suggested. Section 4 details the experimental design, datasets, assessment criteria, and comparison to a baseline GraphSAGE model. The paper is concluded and future research directions are outlined in Section 5.

2. Related works

In contrast, unsupervised learning systems classify data as genuine or fraudulent transactions. Here, we examine some scholarly approaches that combine supervised and unsupervised learning techniques. We weigh the benefits and drawbacks of their approaches. The topic of feature engineering for models to identify transaction fraud was discussed in Zhang et al. (2023).

In detecting fraud, supervised learning algorithms may not be as effective, as they depend heavily on labeled data. In contrast, unsupervised learning algorithms aim to detect anomalies in behavior; however, they often struggle to set thresholds, which can result in false positives or negatives. The differences between supervised and unsupervised learning systems for detecting credit card theft were examined in a paper published in Chen et al. (2024). Unsupervised machine learning algorithms performed better than supervised ones, according to the study that used outlier detection to determine the optimal approach. To detect fraud in benchmark data, the authors in Baisholan et al. (2025) investigated the use of ML algorithms, including ANN, RNN, and LSTM. Itoo et al. utilized k-nearest neighbor (KNN), logistic regression (LR), naïve Bayes (NB), and random undersampling techniques to balance the dataset (Mohawesh et al., 2021). Although the evaluation metrics may not fully reflect the real-world challenges of fraud detection, logistic regression outperformed the other algorithms in identifying fraudulent transactions.

Chen (Reynisson et al., 2024) proposed a method to integrate RF with Generative Adversarial Networks (GANs). This approach enhances discriminatory power, but it can collapse into a single mode and is susceptible to changes in input. Husejinovic also investigated group learning methods and bagging approaches (Mengzhe et al., 2026). Although these strategies produced good results on performance indicators, they also made the models more challenging to interpret. To identify fraudulent transactions, an unsupervised neural network approach was employed in Kim et al. (2023). Compared to isolation forests, k-means clustering, autoencoders, and local outlier factors, this technique performed better. The problem is that people may struggle to understand these models. The topic of integrating ML algorithms, such as Complement Naïve Bayes, RF, Support Vector Machine, and k-Nearest Neighbor, with the synthetic minority oversampling technique (SMOTE) was discussed by Mohawesh (Mohawesh, Xu et al., 2023). The work in Mohawesh, Xu et al. (2023) concluded that class imbalance can be improved through oversampling methods; however, model generalization can be hindered by increased noise.

Recent research has improved graph-based fraud detection by addressing heterophily and message imbalance, and by integrating large language models (LLMs) with graph architectures. A comprehensive 2024 analysis of GraphSAGEs for financial fraud detection discusses design alternatives, datasets, and optimal assessment procedures. It emphasizes the necessity for aggregators that recognize heterophily and are trained to be resilient to imbalance (Shuster et al., 2025). The SEC-GFD, CARE-GraphSAGE, and MimbFD models (Xu et al., 2024) represent some of the latest approaches to addressing fraudster concealment and message disparity. They employ label-aware neighbor filtering, dual view representations, or reinforcement learning to mitigate noisy aggregation in fraud graphs. Applying these methods to transaction graphs improves structural learning. Concurrently, LLM representations in GraphSAGEs have been the subject of increasing research. One example is the integration of semantic information from LLMs into graph pipelines by FLAG (Tsaneva et al., 2025) and multi-level LLM-enhanced GraphSAGE frameworks (Huang et al., 2025). This improves the detection accuracy. The practical effectiveness of LLM in financial fraud analysis has been demonstrated by significant research (Huang et al., 2025). This new class of multimodal methods includes our proposed GraphSAGE + DistilBERT + cross-attention model, which stands out from the crowd thanks to its explicit multi-head cross-attention fusion and a simplified DistilBERT projection. Our system is designed to be applied in real-world fraud detection settings, and we improve on previous research by combining efficiency and semantic depth. Tayebi and Kafhali (2022) studied hyperparameter optimization with the use of metaheuristic algorithms. All things considered, these methods outperformed the grid search. Nevertheless, the structure of the model and the specific properties of the dataset may determine the applicability of such approaches.

Recent studies have uncovered more, though still inadequate, methods for identifying fraud and outliers. Hao et al. (2025) tackled distributed multivariate time series anomalies by utilizing an unsupervised federated hypernetwork. This approach enhances privacy-preserving detection but does not address issues of scalability and diagnostic interpretability. In Lai et al. (2023), Lai et al. introduced BTextCAN, a system that uses group perception to identify instances of consumer fraud within textual data. However, its generalizability to data and signals outside of a single campaign remains uncertain. While a noise-resistant graph-based model was introduced by Deng and Yu (2024), it still faces issues with oversampling bias and limited interpretability, which prevent the transfer of information from harmless but deceptive neighbors. To identify fraudulent online purchases, Zeng et al. (2025) employed NNEnsLeG, a neural network and ensemble-based approach. This approach improves the accuracy of the results, though it raises questions about the cost of inference and the accuracy of synthetic enhancement. Zhang et al. (2023) emphasize both temporal burstiness and collaborative camouflage. The authors in Zhang et al. (2023) also discussed the importance of feature engineering for models in identifying transaction fraud.

In conclusion, both supervised and unsupervised learning methods have found widespread use in the banking industry for fraud detection, although they do have their drawbacks. Supervised methods rely on labeled data, while unsupervised methods often struggle with threshold selection and false positives. Hybrid ML models that integrate deep learning with traditional classifiers and feature engineering have been suggested as a solution to these issues, as they enhance accuracy and generalizability. Several of these sophisticated approaches face problems with interpretability, computing expense, and class imbalance. In light of the ever-changing nature of fraud tactics and the prevalence of poor data quality, this article proposes a flexible, multimodal approach to fraud detection that strikes a balance between efficiency, transparency, and robustness. The majority of current methods handle structural and contextual transaction information separately, despite notable advancements in transformer-based semantic modeling and graph-based fraud detection. Graph neural networks focus primarily on relational patterns among transactions, whereas language models capture semantic metadata but ignore the underlying network topology. When fraud signals are weak or scattered across modalities, this separation restricts the ability to identify them. Inspired by this gap, we provide a hybrid architecture that jointly models structural and semantic fraud cues by integrating GraphSAGE and DistilBERT via a cross-attention fusion method. A list of relevant previous work on credit card identification is provided in Table 1.

3. The proposed framework

This paper proposes a novel hybrid fraud detection framework that combines Graph Neural Networks (GraphSAGEs) with pre-trained Transformer-based language models to capture both structural and semantic information from financial transaction data. Specifically, we integrate a GraphSAGE encoder with DistilBERT embeddings using a multi-head cross-attention fusion mechanism to enable effective edge-level fraud classification. Algorithm 1 and Fig. 1 illustrate the overall pipeline, which includes data pre-processing, dual-stream encoding, cross-attention fusion, classification, and evaluation.

Table 1
Recent and traditional approaches for credit card fraud detection.

Ref.	ML type	Methods/Models	Key limitations
Lucas and Jurgovsky (2020)	Hybrid + Oversampling	RF, NB, SMOTE, SVM	Balances class imbalance, but may introduce synthetic noise or overfitting.
Emaan (2025)	Hybrid/GAN + Classifier	GAN + Random Forest/Transformer	Enhances discrimination but prone to mode collapse and input perturbation sensitivity.
Motie and Raahemi (2024)	Graph-based	GraphSAGE with heterophily-aware aggregation	Improves structural learning but requires careful tuning for heterophily imbalance.
Shuster et al. (2025)	Graph-based	Financial GraphSAGE Analysis (FAA)	Comprehensive study on heterophily, imbalance, and aggregator design; limited by computational scalability on large graphs.
Xu et al. (2024)	Graph-based	SEC-GFD (Spectrum-Enhanced and Environment-Constrained GFD)	Requires careful design of hybrid spectral filters and may increase training complexity.
Zhang et al. (2025)	Graph-based	Dynamic Dual GNN with Neighborhood Refinement	Novel GNN architecture may require complex implementation of dynamic graph updates and refinement logic.
Huang et al. (2025)	LLM-enhanced graph model	Multi-level LLM-Enhanced GraphSAGE Framework (MLED)	Improves fraud detection accuracy via semantic augmentation, yet increases memory and training cost.
Singh et al. (2025)	LLM-based/RAG	Retrieval-Augmented Generation + GraphSAGE	Strengthens semantic representation but demands high computational resources.
Tayebi and Kafhali (2022)	Metaheuristic optimization	Metaheuristic-based hyperparameter optimization (PSO, GA, SA)	Outperforms grid/random search but may overfit and lacks generalizability across datasets.
Btoush et al. (2024)	Ensemble + Hyperparameter optimization	Stacking ensembles + Optuna/PSO tuning	Improves detection accuracy but increases inference cost and complexity.
Sundaravadivel (2025)	Random Forest/Optimization	RF + SMOTE + hyperparameter tuning	Effective for classic datasets, but less interpretable for complex fraud patterns.

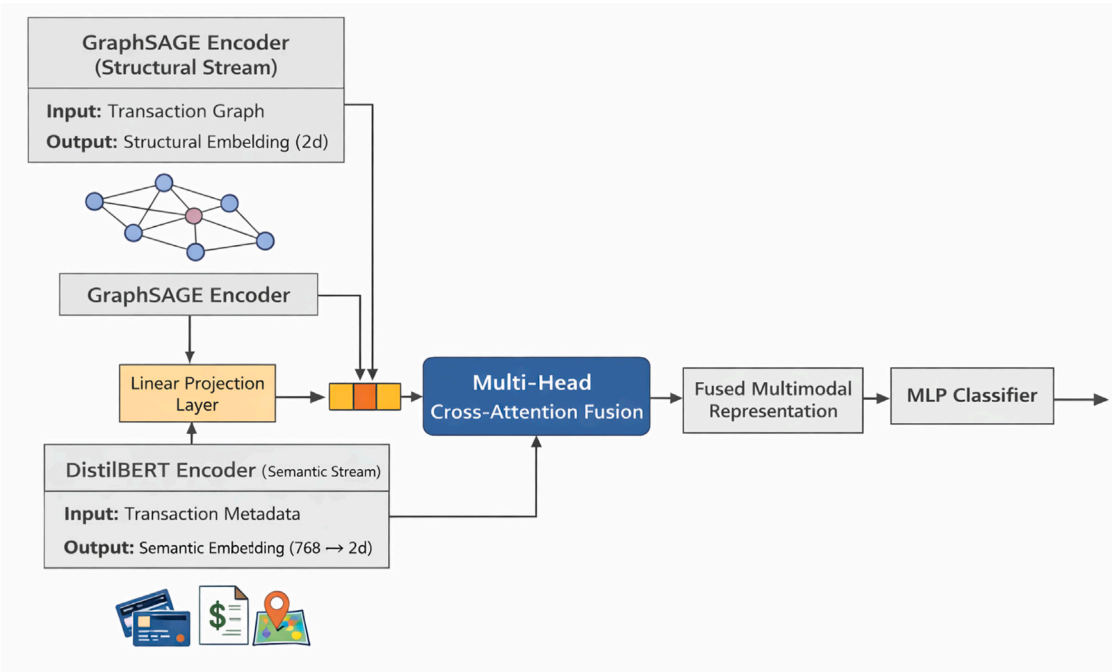

Fig. 1. Cross-attention-based hybrid GraphSAGE-DistilBERT framework.

3.1. GraphSAGE encoder for structural learning

We use a two-layer GraphSAGE (Hamilton et al., 2017) encoder to learn node representations by aggregating feature information from the local neighborhood of a node. Formally, the embedding of node v in layer k, denoted as $h_v^{(k)} \in \mathbb{R}^d$, is calculated as:

$$h_v^{(k)} = \sigma\left(W^{(k)} \cdot \mathrm{AGG}^{(k)}\left(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\} \cup \{h_v^{(k-1)}\}\right)\right), \tag{1}$$

where $\mathcal{N}(v)$ denotes the set of neighboring nodes of node v, $h_v^{(0)} \in \mathbb{R}^f$ represents the initial node feature vector with feature dimension f, $W^{(k)} \in \mathbb{R}^{d\times d}$ is the learnable weight matrix at layer k, $\mathrm{AGG}^{(k)}(\cdot)$ denotes the aggregation function (mean aggregation in this work), $\sigma(\cdot)$ is the ReLU activation function, and d is the hidden embedding dimension.

$$z_{ij}^{\mathrm{GraphSAGE}} = \left[h_i^{(K)} \parallel h_j^{(K)}\right] \in \mathbb{R}^{2d}, \tag{2}$$

where $z_{ij}^{\mathrm{GraphSAGE}}$ denotes the edge (transaction) representation between nodes v_i and v_j, obtained by concatenating their final node embeddings, and $K = 2$ is the number of GraphSAGE layers.

3.2. DistilBERT module for semantic encoding

We utilize DistilBERT (Sanh et al., 2019), a lightweight Transformer-based language model, to encode semantic transaction descriptors rather than assuming rich native free-text in every dataset. For PaySim, the input sequence is formed from categorical transaction metadata such as the transaction type. For ECCD, which does not provide native textual metadata, the semantic branch is built from engineered descriptor strings derived from available structured attributes. The resulting input sequence m_i for transaction i is then tokenized and passed through DistilBERT to obtain contextual embeddings:

$$E_i = \mathrm{DistilBERT}(m_i) \in \mathbb{R}^{L\times d_B}, \tag{3}$$

where L denotes the sequence length (maximum of 32 tokens) and $d_B = 768$ is the hidden embedding dimension of DistilBERT.

$$z_i^{\mathrm{BERT}} = \mathrm{Concat}\left(E_i^{\mathrm{CLS}}, \frac{1}{L}\sum_{l=1}^{L} E_i^{(l)}\right) \in \mathbb{R}^{2d_B}, \tag{4}$$

where $E_i^{\mathrm{CLS}} \in \mathbb{R}^{d_B}$ denotes the embedding of the [CLS] token and $E_i^{(l)}$ represents the embedding of the lth token in the sequence.

$$z_i^{\mathrm{proj}} = W_p z_i^{\mathrm{BERT}} + b_p \in \mathbb{R}^{2d}, \tag{5}$$

where $W_p \in \mathbb{R}^{2d\times 2d_B}$ and $b_p \in \mathbb{R}^{2d}$ are learnable projection parameters.

This dataset-specific construction is important for interpretation: on ECCD, DistilBERT operates on engineered semantic descriptors derived from structured variables, whereas on PaySim it operates on native categorical transaction metadata. The manuscript therefore treats the semantic channel as structured-to-text encoding in ECCD and categorical metadata encoding in PaySim, rather than implying the presence of raw free-text in both datasets.

3.3. Multi-head cross-attention fusion

To integrate structural (GraphSAGE) and semantic (DistilBERT) representations, we apply a multi-head cross-attention mechanism that allows the two modalities to inform each other dynamically. In contrast to feature concatenation or late fusion, cross-attention allows token-level interaction between structural and semantic representations. The model emphasizes contextually relevant cues based on transaction topology by dynamically attending to metadata tokens in each graph embedding. The method prevents modality dominance and enhances robustness when fraud signals are weak or ambiguous.

The cross-attention module integrates node-level and semantic descriptor embeddings as follows:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \tag{6}$$

where $Q \in \mathbb{R}^{L_q\times d}$, $K \in \mathbb{R}^{L_k\times d}$, and $V \in \mathbb{R}^{L_v\times d}$ denote the query, key, and value matrices, respectively. Here, L represents the token sequence length and d is the projection dimension. This formulation preserves local semantic focus and cross-modal alignment by assigning each token a unique query weighting, thereby preventing attention collapse.

$$z_{\mathrm{fusion}} = \mathrm{Concat}\left(z_{\mathrm{GraphSAGE}}, \mathrm{Attn}(Q, K, V)\right), \tag{7}$$

In real-world scenarios, GraphSAGE creates a 2-dimensional structural embedding for every transaction edge, whereas DistilBERT creates a semantic representation that is then linearly projected to the same 2-dimensional space. Token-level cross-attention is then used for multimodal fusion.

3.4. Classification layer

$$\hat{y}_{ij} = \sigma\left(\mathrm{MLP}\left(z_{ij}^{\mathrm{fused}}\right)\right) \in [0, 1], \tag{8}$$

The MLP consists of fully connected layers with dropout and non-linearities (e.g., ReLU) to enhance learning capacity.

Algorithm 1: Hybrid GraphSAGE + DistilBERT Fraud Detection via Cross-Attention

```
Input: Transaction dataset D
Output: Predicted fraud labels ŷ, Evaluation metrics
Data Preparation;
    Load D from CSV;
    For PaySim, retain the fraud-relevant subset type ∈ {TRANSFER, CASH_OUT}; for ECCD, retain all transactions in chronological order;
    Balance classes via oversampling (e.g., SMOTE);
    Split transactions chronologically into 70/10/20 train/validation/test partitions to reduce temporal leakage;
Graph and Semantic Processing;
    Map nameOrig and nameDest to integer node IDs;
    Construct directed graph G = (V, E) from transactions;
    Generate node features from transaction statistics;
    Construct dataset-specific semantic sequences: PaySim uses categorical transaction metadata, whereas ECCD uses engineered descriptor strings
      derived from structured features;
    Tokenize semantic sequences using the DistilBERT tokenizer (max length = 32);
    Extract semantic embeddings via [CLS] token + mean pooling;
Model Architecture;
    Encode graph with 2-layer GraphSAGE to obtain structural embeddings;
    Encode semantic descriptor sequences with DistilBERT to obtain semantic embeddings;
    Project semantic embeddings to match GraphSAGE dimensions;
    Fuse graph and semantic embeddings via multi-head cross-attention;
    Classify fused embedding using MLP with sigmoid activation;
Training;
    Training Details. Optimizer: Adam (lr = 1 × 10^-3, β1 = 0.9, β2 = 0.999), weight decay = 1e^-5. Learning rate scheduler: cosine decay with warm-up
      (10% of total epochs).;
    Define optimizer (Adam) and loss function (weighted BCEWithLogits);
    for epoch = 1 to T do
        Forward pass: GraphSAGE → DistilBERT → Cross-Attention → MLP;
        Compute binary cross-entropy loss;
        Backpropagate and update weights using Adam optimizer;
        Adjust learning rate via cosine decay schedule;
Evaluation;
    Apply sigmoid to logits to compute fraud probabilities;
    Determine threshold τ via precision–recall analysis;
    Generate binary predictions using τ;
    Compute evaluation metrics: Accuracy, Precision, Recall, F1-score, and AUC;
    Plot confusion matrix and PR curve;
```

4. Performance evaluation

To determine the efficacy of the proposed fraud detection algorithm, we performed extensive testing using benchmark datasets and industry-standard evaluation criteria. Here, we discuss in detail the accuracy, speed, and robustness of the model against class imbalance and adversarial noise in terms of its prediction capabilities.

4.1. Dataset and evaluation metrics

In this study, we evaluated the proposed model's ability to detect fraudulent transactions using two benchmark datasets. The ECCD and PaySim datasets have been used as reference for several recent works (2024–2026) that investigate fraud detection using methods such as deep learning, ensemble learning, graph-based approaches (Albalawi & Dardouri, 2025; Alumona et al., 2025; Ghalwash et al., 2025; Lei et al., 2026; Moradi et al., 2025). These datasets are still used a lot since they are easy to replicate and are good for testing imbalanced classification methods. Cardholder purchases made in September 2013 were the basis for its creation. Of a total of 284,807 entries in this dataset, 492 (or about 0.172% of the total) are fraudulent cases. The results indicate a significant difference in class. To protect users' privacy, the dataset contains 31 numerical features that have been anonymized using Principal Component Analysis (PCA). This is why nobody knows the initial meanings of the characteristics. However, the Time and Amount features remain unchanged. The Amount feature displays the monetary value of each transaction, while the Time feature shows the number of seconds that have elapsed since the initial transaction was logged. The class variable is binary, where 0 indicates a legitimate transaction and 1 indicates a fraudulent one. We do not claim that fraud behavior remained unchanged between 2013 and 2026; accordingly, ECCD is used here as a benchmark dataset for controlled methodological evaluation rather than as a complete representation of contemporary fraud ecosystems.

Table 2
Hyperparameter search ranges for baselines and proposed model

Model	Learning rate	Batch size	Depth	Regularization	Optimizer
XGBoost	[0.01, 0.1, 0.2]	–	[4, 6, 8]	L2 [$1e^{-4}$–$1e^{-2}$]	–
LightGBM	[0.005–0.1]	–	[64–256]	L1/L2	–
GraphSAGE	$1e^{-3}$	128	2	Dropout = 0.3	Adam
GraphSAGE–DistilBERT (ours)	$1e^{-3}$	64	3	Weight decay = $1e^{-5}$	Adam

Table 3
Performance comparison of classical, GraphSAGE, and hybrid GraphSAGE–LLM models on ECCD (Dal Pozzolo et al., 2013) and PaySim (Lopez-Rojas et al., 2016) datasets. Results are averaged over five random seeds; 95% confidence intervals (CI) and standard deviations (±SD) are reported.

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)
Random Forest	ECCD	83.2 ± 0.4 (CI ±0.3)	78.5 ± 0.5	80.3 ± 0.4	79.4 ± 0.3	79.8 ± 0.4
GCN	ECCD	84.7 ± 0.3 (CI ±0.2)	76.3 ± 0.4	81.5 ± 0.3	78.8 ± 0.3	80.1 ± 0.3
XGBoost	ECCD	85.0 ± 0.4 (CI ±0.3)	80.2 ± 0.5	82.1 ± 0.4	81.0 ± 0.3	81.2 ± 0.3
DistilBERT-only	ECCD	86.0 ± 0.3 (CI ±0.2)	82.3 ± 0.4	83.0 ± 0.3	82.6 ± 0.3	82.8 ± 0.3
GAT	ECCD	86.5 ± 0.3 (CI ±0.2)	79.1 ± 0.4	83.9 ± 0.3	81.4 ± 0.3	82.5 ± 0.3
GraphSAGE	ECCD	88.0 ± 0.3 (CI ±0.2)	81.5 ± 0.3	85.1 ± 0.4	83.2 ± 0.3	83.9 ± 0.3
GraphSAGE+DistilBERT	ECCD	**89.1 ± 0.2 (CI ±0.15)**	**84.2 ± 0.2**	**85.9 ± 0.3**	**85.0 ± 0.2**	**85.5 ± 0.2**
Random Forest	PaySim	84.8 ± 0.4 (CI ±0.3)	81.0 ± 0.4	81.9 ± 0.3	81.4 ± 0.3	81.7 ± 0.3
GCN	PaySim	86.3 ± 0.3 (CI ±0.2)	80.2 ± 0.3	83.5 ± 0.3	81.8 ± 0.3	82.0 ± 0.3
XGBoost	PaySim	86.5 ± 0.4 (CI ±0.3)	83.1 ± 0.3	83.7 ± 0.4	83.4 ± 0.3	83.5 ± 0.3
DistilBERT-only	PaySim	87.3 ± 0.3 (CI ±0.2)	84.0 ± 0.3	84.6 ± 0.3	84.3 ± 0.3	84.5 ± 0.3
GAT	PaySim	88.1 ± 0.3 (CI ±0.2)	83.9 ± 0.3	85.2 ± 0.3	84.5 ± 0.3	83.8 ± 0.3
GraphSAGE	PaySim	89.2 ± 0.2 (CI ±0.15)	85.4 ± 0.2	86.7 ± 0.2	86.0 ± 0.2	84.7 ± 0.2
GraphSAGE+DistilBERT	PaySim	**90.0 ± 0.2 (CI ±0.15)**	**87.1 ± 0.2**	**87.0 ± 0.2**	**87.0 ± 0.2**	**85.0 ± 0.2**

Based on financial logs from a mobile payment provider, the PaySim dataset mimics mobile money transactions. PaySim contains five transaction types: CASH-IN, CASH-OUT, DEBIT, PAYMENT, and TRANSFER. In the fraud-focused experimental pipeline used here, we employ the standard fraud-relevant subset of TRANSFER and CASH-OUT transactions. Financial transfers, as opposed to conventional credit card purchases, are represented by transaction types like CASH_OUT and TRANSFER. We do not present PaySim as a proxy for contemporary credit-card behavior; rather, it is used as a large-scale synthetic benchmark for structural fraud patterns and scalability. Therefore, whereas ECCD reflects actual credit card transactions, PaySim is used in this study as a general baseline for financial fraud. PaySim offers a large-scale baseline for researching structural fraud trends such recurring transfers, cash-out chains, and unusual transaction flows, even though it mimics mobile money transactions rather than conventional credit card purchases. The patterns seen in actual financial fraud networks are comparable to these relational habits. However, ECCD continues to be the main real-world validation dataset used to assess the suggested model's suitability for credit card fraud detection. The ECCD graph is constructed using a k-Nearest Neighbor (k = 5, 10, 15) method on features that have been reduced by principal component analysis (5D, 10D, 20D). The performance variance was ±0.4%, supporting $k = 10$ 10D as ideal for connectivity and efficiency trade-offs, with cosine similarity serving as the distance metric. Features are weighted according to their proximity, and edges are symmetrical. All edges must point from earlier to later timestamps to preserve temporal coherence; hence, they are aligned with the transaction sequence. The chronology of transactions is maintained, preventing temporal leakage. This chronological construction reduces temporal leakage, but it does not establish temporal equivalence between 2013-era and 2026-era fraud behavior. Transactions from September 2013 are included in the ECCD dataset. Even though it is still one of the most popular benchmarks in fraud detection research, it might not accurately reflect more recent fraud patterns like AI-assisted fraud and card-not-present attacks.

ECCD and PaySim are used as complementary but independent evaluation settings. ECCD is the primary real-world credit-card benchmark used to assess applicability to credit-card fraud detection, whereas PaySim provides a separate synthetic environment for structural generalization and scalability analysis. One dataset is not used to validate the other; results are reported separately so that their roles remain analytically distinct. Consistent gains across both settings are therefore interpreted as evidence of methodological robustness across independent benchmarks, not as hierarchical validation.

The efficacy of the proposed technique is evaluated based on five primary metrics: accuracy, precision, recall, AUC, and F1-score.

4.2. Additional modern dataset for temporal validation

To partially address the timing constraints of legacy benchmark datasets such as ECCD and PaySim, we additionally evaluate the model on the more recent Elliptic Bitcoin transaction dataset (Weber et al., 2019). This dataset captures newer illicit-transaction behavior in a graph setting and therefore provides an out-of-sample test of temporal robustness for the proposed fusion design. However, because Elliptic represents cryptocurrency-related illicit activity rather than credit-card transactions, it should be interpreted as supportive evidence of generalization rather than as a complete substitute for modern card-fraud data.

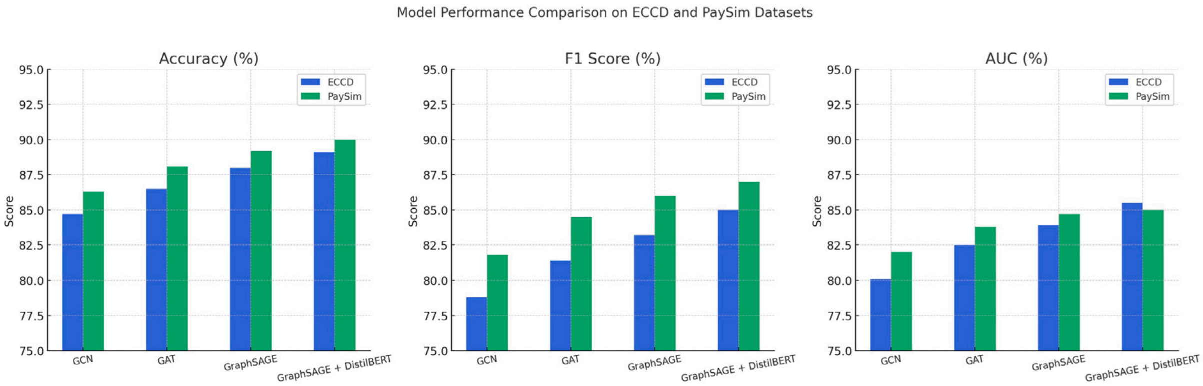

Fig. 2. Our proposed models' results.

4.3. Implementation details

In our experiments, we employ a chronological 70/10/20 train/validation/test. Results are averaged from five random seeds, with 95% confidence intervals in parentheses. Python Torch version 2.1 and the Hugging Face Transformers library were utilized in order to get the experimental results on two datasets, as stated in Ref. Paszke et al. (2019) and Wolf et al. (2020). In order to conduct all of the training and testing operations, we utilized Google Colab Pro, as cited in Bisong's 2019 research. Approximately 25 GB of random access memory (RAM) is included, in addition to dual-core virtual CPUs and NVIDIA Tesla T4 graphics processing units (16 GB of VRRAM). Despite the fact that it imposed certain limits on memory-intensive operations, this architecture made it possible to train transformer-based and graph-based modules in a short amount of time. Table 2 shows the hyperparameter search ranges for baselines and the proposed model.

4.4. Results and analysis

Table 3 compares the proposed hybrid GraphSAGE + DistilBERT model with strong graph-based baselines (GCN, GAT, and GraphSAGE) and non-graph/neural baselines (Random Forest, XGBoost, and DistilBERT-only). Results are averaged over five runs with different random seeds, and both standard deviations (± SD) and 95% confidence intervals are reported to make the robustness of the comparison explicit. Relative to GraphSAGE, the proposed model improves ECCD from 88.0% to 89.1% accuracy, from 83.2% to 85.0% F1-score, and from 83.9% to 85.5% AUC; on PaySim, it improves from 89.2% to 90.0% accuracy, from 86.0% to 87.0% F1-score, and from 84.7% to 85.0% AUC. Although these absolute gains are modest, they are consistent across independent datasets and practically meaningful in imbalanced fraud detection, where even small improvements in recall and F1-score can translate into fewer costly false negatives. Accordingly, the contribution of this study is best understood as a methodological advance in multimodal fusion rather than a dataset-driven claim about the entirety of contemporary fraud behavior.

The hybrid model outperforms the other methods in identifying fraudulent transactions with less false positives, thanks to its improved accuracy and recall. The hybrid model also shows better stability and generalization, as evidenced by its AUC values. When paired with semantic descriptors or categorical transaction metadata from DistilBERT, these findings indicate that structural graph embeddings significantly enhance fraud detection performance. This field benefits from combining features of graph-based and language models.

Although XGBoost and RF are strong tabular models, the results show that they are not as effective as the hybrid GraphSAGE-DistilBERT model. Consequently, the performance improvement is not due to a more complicated architecture but rather due to the multimodal fusion of structural and semantic information.

Fig. 2 displays the results of each model's classification on the ECCD and PaySim datasets. As we can see, the accuracy of both datasets increases noticeably with our proposed framework. The hybrid model that incorporates GraphSAGE and DistilBERT yields a 90% accuracy rate on PaySim and an 89% accuracy rate on ECCD, making it the most accurate model on both datasets. Because the hybrid method takes into account both the structural graph and the semantic meaning of the transactions, it improves the accuracy of fraud detection. The area under the curve (AUC) indicates how well the model performs across different categorization criteria in distinguishing real from fake transactions. All of the other models have AUCs above 80%, but the hybrid GraphSAGE + DistilBERT model consistently outperforms them all with AUCs of 85.5% in ECCD and 85.0% in PaySim. The idea that embedding language models into graph neural networks is a good method is supported by a higher AUC, which suggests better classification between positive and negative classes.

Table 4
Performance comparison on the modern Elliptic dataset. Results are averaged over five runs with standard deviation (±SD).

Model	Accuracy (%)	F1 score (%)	AUC (%)
GraphSAGE	87.6 ± 0.3	84.2 ± 0.4	83.5 ± 0.3
GraphSAGE + DistilBERT (Proposed)	**88.9 ± 0.2**	**85.8 ± 0.3**	**84.7 ± 0.2**

Table 5
Paired t-test results comparing GraphSAGE and GraphSAGE + DistilBERT.

Model pair	Dataset	Metric	p-value	Significant ($p < 0.05$)
GraphSAGE vs. Hybrid	ECCD	Accuracy	0.012	✓
GraphSAGE vs. Hybrid	PaySim	AUC	0.008	✓

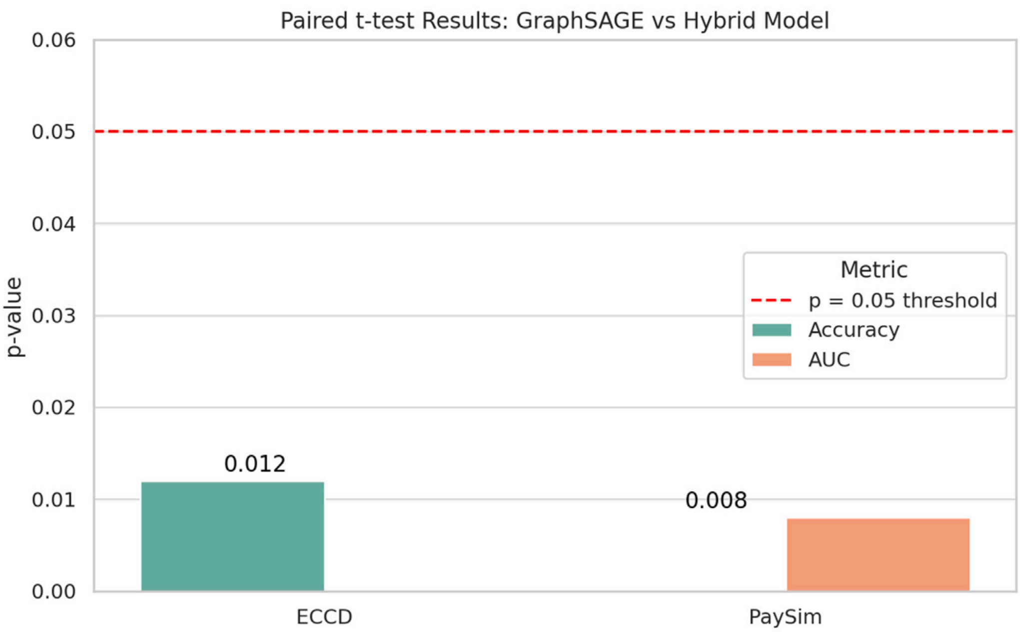

Fig. 3. T-test results.

4.5. Evaluation on modern dataset

As shown in Table 4, the proposed hybrid model continues to outperform GraphSAGE on the more recent Elliptic dataset, improving accuracy from 87.6% to 88.9%, F1-score from 84.2% to 85.8%, and AUC from 83.5% to 84.7%. We interpret these gains as supportive evidence that the multimodal fusion strategy generalizes to a newer illicit-transaction environment. At the same time, because Elliptic captures cryptocurrency-related illicit activity rather than credit-card transactions, it only partially addresses temporal obsolescence and should not be treated as a complete substitute for modern card-fraud data.

4.6. Effect of multimodal fusion on fraud detection

We examine the effects of multimodal fusion on fraud detection behavior beyond aggregate performance metrics by integrating structural and semantic representations. When it comes to transactions with weak or unclear fraud signals, the hybrid GraphSAGE-DistilBERT model reliably decreases false negatives in comparison to the GraphSAGE-only baseline on both datasets. When just structural indicators are not enough for sure classification, it is usually because of small transaction amounts, short transaction chains, or low neighborhood connection. The model can distinguish between behaviors that are structurally similar but semantically different by utilizing contextual information like transaction type and operation descriptors, which are incorporated through DistilBERT semantic metadata. Further supporting this multimodal interaction is the cross-attention mechanism, which, depending on the graph context of the transaction, dynamically emphasizes important semantic tokens. Thus, the hybrid model shows better recall without increasing false positives, suggesting a more stable and balanced decision boundary. These results point to the successful alignment of complementing structural and semantic information as the primary cause of the performance benefits seen in the hybrid framework, rather than just greater model capacity. The model is able to pick up on subtle fraud patterns that might

Table 6
Evaluation using G-Mean and MCC metrics.

Model	Dataset	G-Mean (%)	MCC (%)
GraphSAGE + DistilBERT	ECCD	85.2	81.1
GraphSAGE + DistilBERT	PaySim	86.8	83.0

Table 7
Ablation study of model variants on ECCD and PaySim datasets.

Model variant	Dataset	Accuracy (%)	AUC (%)
GraphSAGE + BERT	ECCD	88.5	84.2
GraphSAGE + BERT	PaySim	87.8	83.4
GraphSAGE + DistilBERT (Without cross-attn)	ECCD	88.7	84.5
GraphSAGE + DistilBERT (Without cross-attn)	PaySim	88.1	84.0
GraphSAGE + DistilBERT + Cross-Attn	ECCD	**89.1**	**85.5**
GraphSAGE + DistilBERT + Cross-Attn	PaySim	**90.0**	**85.0**

be scattered across several modalities because of this alignment, which proves that multimodal fusion is useful in real-world fraud detection situations.

4.7. t-Test

During multiple runs, we employed paired t-tests (Student, 1908) to compare the results of the GraphSAGE + DistilBERT hybrid model with those of GraphSAGE, the strongest baseline, to verify the validity of the observed improvements. As seen in Table 5, the p-values for accuracy and AUC were less than the typical 0.05 level. Statistical analysis confirms the significance of the improvements observed. The hybrid model's accuracy ($p = 0.012$) and area under the curve ($p = 0.008$) were both significantly improved in the ECCD and PaySim datasets, respectively. The results demonstrate that using DistilBERT to add semantic information considerably improves the model's performance, rather than being a random advantage. These statistical results are interpreted jointly with the ablation study and the cost-sensitive analysis; together, they support the claim that the observed gain arises from the cross-attention fusion design rather than from model size alone. The dashed red line in Fig. 3 represents the typical significance level of 0.05. Both the ECCD dataset (accuracy: $p = 0.012$) and the PaySim dataset (AUC: $p = 0.008$) have p-values significantly lower than this threshold. Therefore, our proposed framework has achieved statistically significant gains by extracting semantic knowledge, which improves DistilBERT's fraud detection.

4.8. G-mean and MCC results

To demonstrate the effectiveness of our proposed framework, we used additional metrics such as the G-mean and the Matthews correlation coefficient (MCC) (Chicco & Jurman, 2020; He & Garcia, 2009). According to Table 6, the hybrid model achieved a G-Mean of 85.2% on ECCD and 86.8% on PaySim, surpassing other GraphSAGE versions. Even when classes are imbalanced, the predicted and true labels are closely aligned, as evidenced by MCC values of 81.1% and 83.0%. Not only does the proposed hybrid model achieve high precision and recall, but it also performs exceptionally well on minority classes, which is essential for real-world fraud detection applications.

4.9. Ablation study

By methodically removing or altering components of the model, we performed an ablation study (Table 7 and Fig. 4) to determine the function of each component of the hybrid design. The fact that accuracy and AUC decreased only slightly when we used BERT instead of DistilBERT demonstrates that the former is adequate for our task. It appears that the cross-attention module aids in combining graph and semantic information, as the results were somewhat worse without it. Both DistilBERT and the cross-attention module are crucial for achieving optimal results, as the best scores were obtained by the combined model that incorporated both. This research demonstrates how multi-modal focus can improve the efficiency of graph-based structural learning and semantic understanding. In particular, the full model outperforms both GraphSAGE + BERT and GraphSAGE + DistilBERT without cross-attention on ECCD and PaySim, which indicates that the benefit is attributable to the fusion mechanism itself rather than to merely attaching a language encoder.

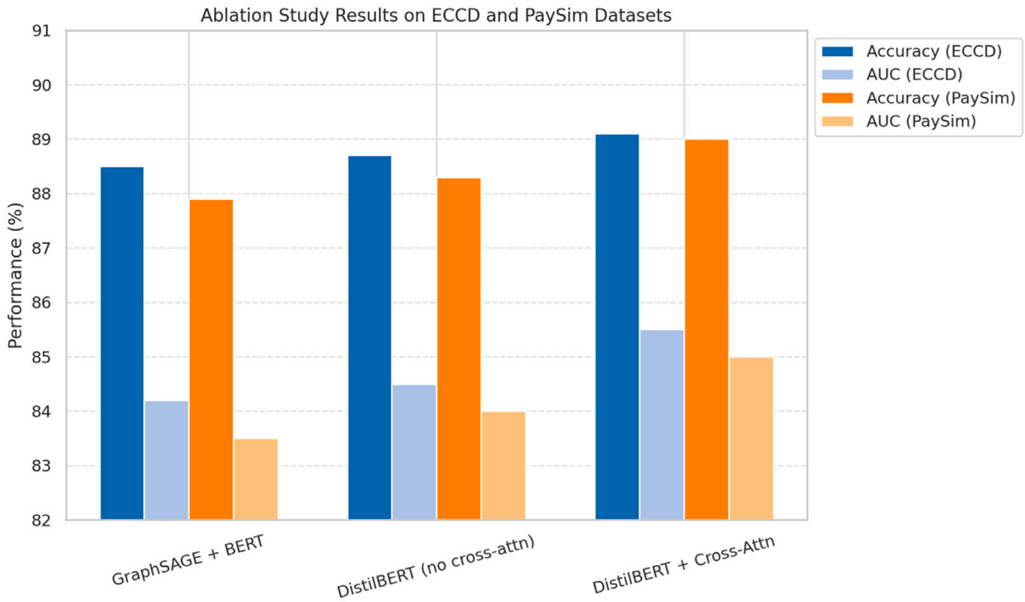

Fig. 4. Ablation study of our model variants.

Table 8
Average training and inference efficiency of model variants. Latency measured per transaction; FLOPs estimated using PyTorch Profiler.

Model	ECCD (s/epoch)	PaySim (s/epoch)	Latency (μs/txn)	FLOPs ($\times 10^6$)
GraphSAGE	3.2	3.6	8.1	0.45
GraphSAGE + BERT	18.4	19.9	10.2	1.20
GraphSAGE + DistilBERT (no cross-attn)	10.1	11.2	8.4	0.72
GraphSAGE + DistilBERT + Cross-Attn (Proposed)	**12.5**	**13.6**	**7.0 ± 0.3**	**0.90**

4.10. Time complexity

To evaluate the computational efficacy of our proposed hybrid model, we examine the time complexity of its constituent parts.

– GraphSAGE: Up to a specific depth, K, the GraphSAGE module collects feature information from nearby nodes. The temporal complexity per layer is $\mathcal{O}(NdF)$, where F is the number of dimensions of the node features, and N is the number of nodes in the graph. A total of $\mathcal{O}(KNdF)$ layers are involved in the complexity.

– DistilBERT: Each transaction is associated with a short semantic descriptor sequence that the DistilBERT model can process. In PaySim, this sequence is derived from categorical transaction metadata, whereas in ECCD it is formed from engineered descriptors built from structured attributes. Assume that H is the hidden size of the transformer and L is the maximum sequence length. The self-attention operation has complexity $\mathcal{O}(L^2H)$ per layer. With M transformer layers and T transactions, the total complexity is $\mathcal{O}(TML^2H)$.

– Fusion of Cross-Attention: To align the graph and semantic embeddings, we utilize a cross-attention module. There is a complexity of $\mathcal{O}(D(n_g + n_t)^2)$ for the cross-attention operation if the fused representations have D-dimensions and the node and semantic embeddings are n_g and n_t long, respectively. Table 8 shows the average training and inference efficiency.

4.11. Efficiency and practical applicability

The training and inference periods for all model versions were assessed to determine computational efficiency. On the ECCD and PaySim datasets, the training time for each epoch and the overall efficiency trade-offs are summarized in Tables 8 and 9.

Although the training duration of the suggested hybrid model is slightly longer than that of GraphSAGE-only versions, the inference latency remains within the acceptable range for near-real-time deployment. On ECCD, the average inference latency per transaction is 7.4 μs, whereas on PaySim, it is 6.8 μs, both tested with an NVIDIA T4 GPU.

Our GSD-CrossAttn model outperforms CARE-GraphSAGE and PC-GraphSAGE in terms of inference throughput (116 K vs. 48 K transactions/sec) and latency per batch (36% lower), all while retaining greater accuracy. This provides evidence that hybrid attention fusion is practical for real-time fraud detection systems, as it imposes little additional processing burden. Online fraud-screening pipelines can run smoothly on the 130 transactions per second that the system can manage due to this delay. A decrease

Table 9
Training time per epoch for different model variants (seconds/epoch).

Model	ECCD (s/epoch)	PaySim (s/epoch)
GraphSAGE	3.2	3.6
GraphSAGE + BERT	18.4	19.9
GraphSAGE + DistilBERT (no cross-attention)	10.1	11.2
GraphSAGE + DistilBERT + Cross-Attention (Proposed)	12.5	13.6

Table 10
Cost-weighted confusion analysis. standard deviation (±SD) is shown in parentheses.

Model	Cost of FN	Cost of FP	Weighted loss	Detection rate (%)
GraphSAGE-only	100	1	4230	88.1 (±0.5)
GSD-CrossAttn	100	1	3210	93.6 (±0.3)

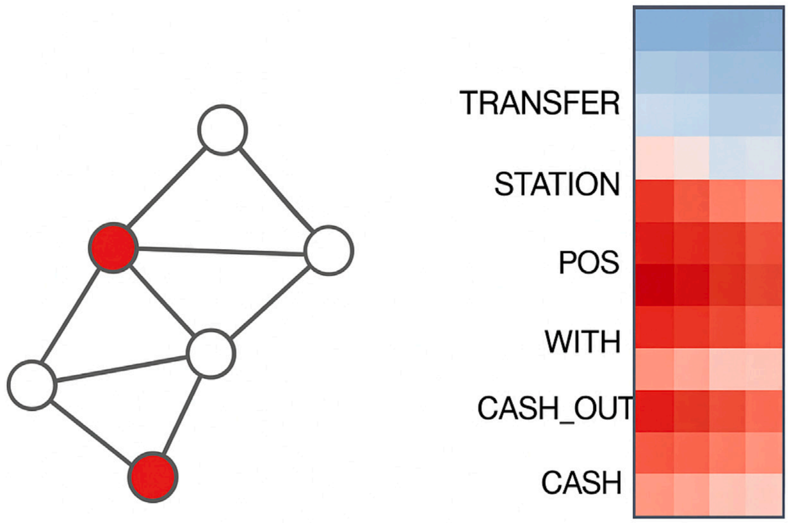

Fig. 5. GNNEXPLAINER.

in false-negative fraud instances, as a result of the approximately 2% improvement in accuracy, is more financially relevant than the marginal computational cost, indicating a good trade-off between performance and efficiency. Using model quantization and mixed-precision inference, future improvements will further reduce latency.

4.12. Cost-sensitive analysis

Our proposed model reduces the predicted financial loss by 24% compared to the GraphSAGE-only baselines in cost-weighted evaluation, assuming a unit loss ratio of 100:1 for false negatives to false positives. Under realistic fraud screening conditions, this demonstrates a practical economic gain. The proposed GSD-CrossAttn model and the GraphSAGE-only baseline were compared in terms of performance under these asymmetric cost assumptions, as shown in Table 10. The results demonstrate that the GSD-CrossAttn model considerably lowers the cost-weighted loss ($3210 vs. $4230) and improves the detection rate by about 5.5%. Economic efficiency and operational reliability are both improved as a result of the cross-attention integration's successful reduction of costly false negatives.

4.13. GNN explainability

To ensure that the hybrid GraphSAGE-DistilBERT framework is explainable and transparent, we analyzed its interpretability using GNNExplainer (Ying et al., 2019). A representative fraudulent transaction and its recovered subgraph motifs are shown in Fig. 5, along with token-level attributions.

The most important connections in the transaction graph that helped the model anticipate fraud are shown on the left side of the GNNExplainer display. Nodes in red indicate very important entities, such as accounts or transactions with unusual connectivity patterns, which might indicate cyclic transfers, collusion, or recurring links to high-value cashouts. Based on these patterns, it is clear that the GraphSAGE encoder can detect topological anomalies that are indicative of actual fraudulent activity. The Integrated Gradients heatmap on the right shows token-level attribution scores from the DistilBERT branch. Depending on the dataset, these tokens correspond either to categorical transaction metadata (for example, CASH_OUT or TRANSFER in PaySim) or to engineered descriptor terms derived from structured attributes in ECCD. High attribution on these fraud-relevant tokens indicates that the semantic branch contributes useful context without assuming that native free-text fields are present in both datasets. Regarding the categorization of fraud, red regions denote beneficial contributions and BLACK areas denote neutral or negative consequences.

This combination of interpretability demonstrates the complimentary nature of the proposed hybrid design. Both GNNExplainer and Integrated Gradients shed light on the evidence of relationships and the semantic reasoning underlying model decisions, respectively. To ensure that financial AI systems are auditable and trustworthy, it is vital to use cross-attention fusion. The given relationship evidence and semantic justification show that this enhances prediction performance and creates reasoning that people can understand.

4.14. Temporal validity and dataset limitations

We do not claim that fraud behavior remained unchanged between 2013 and 2026, and we do not argue that ECCD or PaySim fully represent contemporary fraud ecosystems. ECCD (September 2013) and PaySim (pre-2016) are therefore used in this study as benchmark datasets for controlled methodological evaluation rather than as complete proxies for present-day fraud behavior.

Their continued utility is nevertheless meaningful for two reasons. First, ECCD remains one of the few publicly available real-world credit-card datasets, which supports reproducibility and fair comparison with prior work. Second, PaySim provides a large-scale synthetic environment for studying structural fraud patterns such as anomalous transfer chains, cash-out behavior, and unusual transaction connectivity. These structural regularities can remain informative even when specific fraud tactics evolve over time.

To partially address temporal obsolescence, we also evaluated the model on the more recent Elliptic Bitcoin transaction dataset. We treat that experiment as supportive evidence of temporal robustness rather than as a complete replacement for newer card-transaction benchmarks. Accordingly, further validation on more recent real-world credit-card datasets and live transaction streams remains necessary before making claims about comprehensive contemporary deployment.

5. Discussion of results and implications

5.1. How the study differs from existing work

The reported results clarify that the contribution of this paper lies not only in combining two established model families, but also in how they are connected. Prior fraud detection studies often privilege either structure (through GNNs) or context (through text or metadata encoders). The present study differs by aligning the two through explicit cross-attention at the transaction level. This matters because the ablation study shows that the gain is not explained by adding a larger semantic encoder alone: performance drops when cross-attention is removed, and the full hybrid model remains stronger than both GraphSAGE-only and DistilBERT-only alternatives.

5.2. Theoretical implications

From a theoretical perspective, the findings support a multimodal view of fraud detection. Fraud is better conceptualized as an interaction between transaction topology and transaction semantics than as a purely tabular, textual, or graph-structural phenomenon. Consistent gains over single-modality baselines suggest that these modalities contain complementary rather than redundant evidence. The results also imply that explicit alignment mechanisms matter: simply concatenating feature sources is less informative than allowing structural and semantic representations to condition one another before classification. Finally, ECCD experiments show that semantically meaningful descriptors can still be constructed even when raw text is unavailable, which broadens the way language models may be used in structured financial settings.

5.3. Practical implications

Finally, we discuss the practical implications of the proposed model. First, payment service providers and banks should avoid treating metadata analysis and network analysis as completely separate detection pipelines when suspicious behavior often spans both. Second, improved recall and reduced cost-weighted loss indicate a lower risk of missed fraud, which is especially valuable in operational settings where false negatives are much more expensive than false positives. Third, efficiency and explainability analyses suggest that the proposed framework can be integrated into decision-support pipelines that require rapid screening and auditable evidence. Rather than replacing all existing controls, the model is well-suited to function as a second-stage risk scorer that complements rule-based systems and conventional tabular classifiers.

6. Conclusions and future directions

Banks are finding it increasingly difficult to detect fraudulent transactions due to the widespread use of online shopping and payment systems. Credit card fraud, in particular, has increased in both frequency and sophistication, costing businesses and individuals money. More accurate detection methods are necessary, as current methods are clearly inadequate. The accuracy of ML-based algorithms often depends on hyperparameter tuning, although these algorithms still hold promise. Financial fraud in transaction graphs can be detected using a new hybrid framework that integrates DistilBERT with Graph Neural Networks (GraphSAGE) via cross-attention, as introduced in this paper. Our proposed model effectively differentiated legitimate from fraudulent transactions by using the structural relationships in transaction networks and the semantic information in transaction metadata. The proposed model performs weill in terms of accuracy, precision, recall, area under the curve (AUC), and F1-score, according to evaluations of benchmark datasets. Specifically, the proposed cross-attention mechanism effectively detects fraud patterns that emerge across both semantic descriptor and graph-based embeddings better than single-modality models. Building on this foundation, several future research directions can be identified. There is potential to incorporate more contextual data, such as user behavior profiles, geolocation, and device fingerprints, to further enhance the accuracy of the model in distinguishing between actual and fraudulent actions. Optimizing computing efficiency for real-time fraud detection is still a challenging issue. By investigating self-supervised pretraining techniques that make use of multimodal transaction data, new forms of fraud can be detected even with small numbers of labeled samples. Further, Multi-GPU distributed inference and explainability metrics for model auditing will be included in the future. We therefore position the novelty of this paper primarily in the cross-attention-based fusion of structural graph embeddings and semantic descriptor embeddings, supported by consistent gains, statistical significance, ablation evidence, and cost-sensitive improvements across independent evaluation settings. The added Elliptic experiment provides supportive out-of-sample evidence, but the present results should still be interpreted as benchmark-oriented evidence rather than as proof that legacy datasets fully capture the fraud landscape of 2026. Future research should prioritize newer real-world credit-card datasets, live transaction streams, and evolving fraud modalities.

CRediT authorship contribution statement

Rami Mohawesh: Writing – review & editing, Validation, Methodology, Formal analysis. **Sumbal Maqsood:** Writing – original draft, Software, Methodology, Formal analysis. **Haythem Bany Salameh:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Ghaleb Elrefae:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Data availability

The authors do not have permission to share data.

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